BT604 11am Wednesday
1-processes involved in recovery of products 10

Recovery of products:

Following cell disruption, soluble proteins are usually separated from cell debris by centrifugation.

The solubility of the salt varies with temperature, so strict temperaturE control is required.

Many alkaloids, antibiotics, steroids and vitamins are recovered by liquid–liquid extraction methods using organic solvents.

Chromatographic techniques

Chromatographic techniques are usually employed for higher-value products.

These methods, normally involving columns of chromatographic media (stationary phase), are used for desalting, concentration and purification of protein preparations.

The order and choice of technique will depend on the particular product, but the following chromatographic parameters should be considered: capacity, recovery andresolving power (selectivity).

Affinity chromatography

Affinity chromatography is a particularly powerful and highly selective purification technique

Ligands are covalently linked to the matrix material, e.g. agarose, via a spacer arm to avoid steric hindrance

The loading capacity of affinity columns is large, as the volume of the sample is unimportant and high resolution can be attained

For industrial-scale operations, it is often necessary to sterilize the chromatographic media in order to comply with various

regulatory requirements.

Gel filtration chromatography:

Gel filtration chromatography essentially involves separation on the basis of molecular size (molecular sieving), although molecular shape can also influence separation performance.

The initial choice of stationary phase material is also a key factor, as some may interact with the target product, e.g. carbohydrate-based matrices interact with glycoprotein's.

High-performance liquid chromatography(HPLC)

High-performance liquid chromatography(HPLC) was originally developed for the separation of organic molecules in non-aqueous solvents, but is now used for proteins in aqueous olution.

The protein to be purified must have an affinity for this ion and binds to it by forming coordination complexes with groups such as the imidazole of histidine residues**.**

Membrane separation techniques:

membrane separation techniques are primarily used for the removal of low molecular weight solutes and inorganic ions from a solution

2-uses of microbial enzymes in industries 5

Commercial microbial enzymes are increasingly replacing conventional chemical catalysts in many industrial processes.

Certain enzymes are not restricted to aqueous environments and can operate in two-phase water–organic solvent systems and in non-aqueous organic media, particularly hydrophobic solvents.

Several thousand tonnes of commercial enzymes are currently produced each year, which have a value in excess of US$1500 million.

Applications:

- Besides their role as aids in traditional processes, ‘bulk’ enzymes are at the centre of many novel processes and biotechnological innovation continues to expand the range of applications.
- Proteases are not the only enzymes used in detergents; since the late 1980s, amylases and lipases have been available for incorporation, e.g. Lipolase from Novo Nordisk.
- A Swiss chemist Jaag, who worked for the detergent company Gebrüder Schnyder, developed a new product containing a bacterial protease, which replaced the animal trypsin that had previously been incorporated into these detergent products.
- Microbial enzymes are predominantly produced by submerged fermentations, although some solid-substrate fermentations are used, particularly for the production of extracellular fungal enzymes

3-yeast as eukaryotic host for heterogeneous gene 5

A preferred alternative strategy for the expression of heterologous eukaryotic genes is often the employment of a suitable eukaryotic host

that will naturally perform any necessary post-translational protein modification and secretion.

- Initially, *saccharomyces cerevisiae was a* popular choice because it is safe and a vast amount of information has been accumulated about its genetics, physiology and performance in industrial fermentation processes.
- Also, it has a relatively rapid growth rate and readily undergoes genetic manipulation,
- but unliken higher eukaryotes (animal and plant cells) this yeast is Easily and cheaply grown on an industrial scale. Several heterologous eukaryotic proteins have been successfully mass produced by *S. Cerevisiae.However, product yields* are relatively low at 1–5% of total protein and some proteins are retained within the periplasm.
- Other yeasts may be better hosts, particularly the methylotrophs, *pichia angusta (formerly hansenula polymorpha) and pichia pastoris, which have a number of advantages* over *S. Cerevisiae.*

- They have strong inducible promoters and are capable of generating post-translational protein modification similar to those performed by human cells.
- Downstream processing is also less problematical as they do not secrete many of their own proteins into the medium.
-

4- STR 5

- Stirred tank reactors (STRs) have mechanically moving
- Agitators or impellers within a baffled cylindrical vessel.
- Baffles are usually flat vertical plates whose width is about one-tenth of the vessel diameter.
- Normally, 4–6 baffle plates are fitted to the inside vessel walls to aid mixing and mass transfer by increasing turbulence, preventing vortex formation and eliminating ‘dead spaces’.
- Strs are the most commonly used vessels and have been adopted for a wide range of fermentation processes.
- Within each vessel the impeller is connected to an external motor which drives the stirrer system. The agitator assembly, including the seal, is often a potential route of contamination.
- To prevent this problem, the shaft has to pass into the fermenter through a set of aseptic seals.
- There are specific regulations regarding the numbers and types of seals.
- For certain fermentations, two or three seals are required to minimize the risk of fermenter contamination, and release of microorganisms and their products into the environment.

5-name 5 GRAS microbes 3

Expression in GRAS-listed organisms provides obvious advantages. Prime candidates for this role are species within three genera of microorganisms, namely

- Bacillus,

- Aspergillusand
  Saccharomyce

Saccharomyces cerevisiae

Streptomyces griseus

brown algae

6-Foam production and its disadvantage 3

- Foaming is largely due to media proteins that become attached to the air–broth interface where they denature to form a stable foam.
- If uncontrolled the foam may block air filters, resulting in the loss of aseptic conditions; the fermenter becomes contaminated and microorganisms are released into the environment.

  Disadvantages:

  If uncontrolled the foam may block air filters, resulting in the loss of aseptic conditions; the fermenter becomes contaminated and microorganisms are released into the environment.

7-factors influencing oxygen transfer 2

- Oxygen transfer is complex, as it involves a phase change from its gaseous phase to the liquid phase, and is influenced by the following factors.

1. The prevailing physical conditions; temperature, pressure and surface area of air/oxygen bubbles;
2. The chemical composition of the medium;
3. The volume of gas introduced per unit reactor volume per unit time;
4. The type of sparger system used to introduce air into the fermenter;
5. The speed of agitation; or
6. A combination of these factors.

8-citric acid uses in industries 2

- Citric acid is widely used in the food industry as an acidulant and flavouring agent in beverages, confectionery and other foods, and in leavening systems for baked goods.
- As a food constituent, its use is unrestricted because it has GRAS status.
- They include roles in maintaining metals in solution for electroplating, as a cleaning and 'pickling' agent for metals, and as a replacement for polyphosphates in the detergent industry, along with several pharmaceutical uses.
  9-define metabolic regulation 2

BT604 Paper
Questions:
1. Metabolic Regulation 2

- Microorganisms naturally have tight control over their metabolism,
- But this often needs to be overcome when they are used in industrial fermentations, e.G. In attempting to over-produce specific metabolites.
- Consequently, the study of metabolic regulation is of special importance in industrial microbiology.
- **REGULATORY ENZYMES**
- Many metabolic pathways contain **regulatory enzymes** whose activity can be adjusted to provide a mechanism for fine tuning the flux of carbon through a pathway.
- Some of these enzymes are **allosteric, usually composed** of several subunits, and do not follow michaelis– menten kinetics.
-

2. Uses of Citric Acid 2
3. Factors for oxygen transfer to the medium of fermenter2
4. Uses of microbial enzymes in industries 3
5. Ways of cell harvesting 3

Broth conditioning
Broth conditioning techniques are mostly used in association with sedimentation and centrifugation for the separation of cells from liquid media. Some organisms naturally flocculate, which can be enhanced by

chemical, physical and biological treatments. Such treatments can also be effective with cells that would not otherwise form flocs

Sedimentation

Sedimentation is extensively used for primary yeast separation in the production of alcoholic beverages, and in waste-water treatment.

Centrifugation

If instead of simply using gravitational force to separate suspended particles, a centrifugal field is applied, the rate of solid–liquid separation is significantly increased and much smaller particles can be separated.

6. Why foam is produced and what problem is caused during fermentation 3

7. 5 microorganisms named classified as GRAS 3

8. Stirred tank reactors 5

9. Yeast as a eukaryotic host in industrial area 5

10. Recovery data processes 10

Bt604 final term

Products produced by recombinant DNA

Products which are produced by recombinant DNA technology(produced by biotechnology)

• Examples:

- Antibiotics
- Vaccines
- GMO's
- Transgenic plants
- Baverages

What you know floc formation

There are two main theories of floc formation.

• The first concerns calcium bridging, where a calcium ion links two cells via negatively charged cell wall components. Support for this theory comes from the observation that flocs are dispersed by the chelating agent ethylenediamine tetraacetic acid (EDTA).

• Secondly, the lectin hypothesis, involving protein–carbohydrate binding, proposes that a surface protein of one cell links to a mannose residue on an adjacent cell. This hypothesis is supported by the fact that sugars, particularly mannose, inhibit floc formation

Describeclassification of microorganisms on the basis of their requirements for oxygen.

Microorganisms are classified into five main groups on the basis of their requirements for oxygen.

• **1 obligate aerobes** grow only in the presence of oxygen, as it is required as the terminal electron acceptor for electron transport in aerobic respiration

• **2 facultative anaerobes** function with or without oxygen, but grow more efficiently when oxygen is available.

3 microaerophiles require some oxygen for the biosynthesis of certain compounds, but cannot grow at normal atmospheric oxygen concentrations of 21% (v/v). They must have lower oxygen levels of 2– 10% (v/v).

• **4 aerotolerant anaerobes** essentially ignore oxygen and grow equally well in its presence or absence.

• **5 obligate anaerobes** cannot tolerate oxygen; exposure to it results in their death.

Enlist microorganism used in fermentation process

Fermentation is a process by which a substance breaks down into a simpler substance.

• Microorganisms like yeast and bacteria usually play a role in the fermentation process, creating beer, wine, bread, kimchi, yogurt and other foods.

• Fermentations use an organic molecule, pyruvate or a derivative, as this final electron acceptor, thereby regenerating NAD(p)+ and allowing catabolism to continue.

5 microorganisms of Gras
Formation of foam
Citric acid

1. **Definition of ascomycete:** This is the largest class of fungi and includes the yeasts that are utilized in
   many industrial fermentation processes
   2. Defi.....of catabolism 1 the break down of complex molecules into simpler ones with the release of energy is
   called catabolism
   3. Defi ....of white 4.biotechnology Also known as industrial biotechnology applies to industrial processes.
   - Designing of an organism to use to produce a useful chemical.

- Use of enzymes as industrial catalysts to produce valuable chemicals or destroy hazardous/ polluting chemicals.
• Started in California in 1970s.

4. E coli discoverd in E. Coli was discovered in 1885 by the German bacteriologist Theodor escherich.

5. In chemical substance bacteria fungi kill by

6. In spred plating ..sample volume is In spread plating, samples, usually 0.1ml, are spread on the surface of a suitable
agar-based nutrient medium using a sterile spreading device, e.G

7. Obligate aerobese need oxygen(2mcq)

8. batch fermentation...

9. Ascomycete 5marks

This is the largest class of fungi and includes the yeasts that are utilized in many industrial fermentation processes
• Neurospora species, claviceps species, and important edible fungi, including morchella species (morels) and tuber species (truffles) are industrially important.
• Their hyphae are septate

Asexual reproduction:

• In asexual reproduction, the tips of specialized hyphae (conidiophores) form chains of haploid conidiospores that are usually wind dispersed.
• **Sexual reproduction:**
• Sexual spores or ascospores are contained within a sac-like ascus, which may be enclosed within a fruiting body or ascocarp

10. Antibiotic 5 marks

Antibiotics may have cidal or static effects on a range of microorganisms.
• They are classified according to their range of effectiveness and the general microbial group they act against, i.E. Antibacterial, antifungal or antiprotozoal.
• The range of bacteria or other microorganisms affected by a specific antibiotic is expressed as its spectrum of action.

2. Antibiotics effective against a wide range of gram-positive and gramnegative bacteria are said to be broad spectrum.
• If they are predominantly effective against gram-positive or gram-negative bacteria, they are narrow spectrum
• Limited spectrum antibiotics are effective against only a single organism or disease.

11. Imperfect fungi

This division contains a diverse group of around 25000 species
• These are grouped together simply because they lack a defined sexual (perfect)

stage

• It is assumed that this group may represent the conidial stages of ascomycetes whose ascus stage has not yet been discovered or has been lost in the course of evolution.

12. Microbial growth

Microbial growth can be defined as an orderly increase in cellular components, resulting in cell enlargement and eventually leading to cell division.

• This definition is not strictly accurate as it implies that a consequence of growth is always an increase in cell numbers.

• However, under certain conditions growth can occur without cell division, In this situation the cell numbers remain constant, but the concentration of biomass continues to increase.

• This is also true for coenocytic organisms, such as some fungi, that are not divided into separate cells.

• Their growth results only in increased size.

13. Note on yeast

14. Metabolic Regulation 2

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• But this often needs to be overcome when they are used in industrial fermentations, e.G. In attempting to over-produce specific metabolites.

• Consequently, the study of metabolic regulation is of special importance in industrial microbiology.

MODIFICATION OF ENZYME ACTIVITY

• This involves the stimulation or inhibition of specific enzymes to rapidly change the flow of carbon within a pathway.

• **Physical separation of enzyme and substrate;** the simplest mechanism for the regulation of enzyme activity merely involves controlling the concentration of available substrate and coenzymes.

• The most obvious mechanism is the control of substrate entry into a cell across the cytoplasmic membrane.

15. Uses of Citric Acid 2

16. Factors for oxygen transfer to the medium of fermenter 2

17. Uses of microbial enzymes in industries 5

18. Ways of cell harvesting 3

19. Why foam is produced and what problem is caused during fermentation 3

20. 5 microorganisms named classified as GRAS 3

21. Stirred tank reactors 5

22. Yeast as a eukaryotic host in industrial area 5

23. Recovery product processes 10
24. Describe microbial growth kinetics..2
25. What is imperfect fungi?..2
26. Explain briefly yeast..3

The term ‘yeast’ refers to a unicellular form rather than to a phylogenetic classification

• It can be used to describe a unicellular phase of the life-cycles of fungi that may be predominantly filamentous, but exhibit yeast–mould dimorphism

• This term is used more commonly as a generic name for fungi that have only a unicellular phaseThese true yeasts and many others of industrial importance are strains of S. Cerevisiae, a member of the ascomycetes

• Yeasts are heterotrophic

• They are found in a wide range of natural habitats, being particularly common aerial plant organs, especially flowers, and plant debris in the surface layers of soil

Many yeasts are facultative anaerobes, able to grow in the absence of oxygen

• Strains of S. Cerevisiae are the most well-known and commercially important yeasts, and have long been employed to produce alcoholic beverages and leaven bread

• They are now also used in the production of several other fermentation products, particularly fuel ethanol, single cell protein, enzymes and heterologous proteins, e.G. Human insulin

27. What is ATP bioluminator..3

The amount of ATP in a sample can be quantified using ATP bioluminometry.

• This technique utilizes an enzyme–substrate complex, luciferase– luciferin, obtained from the firefly, photinus pyralis, which generates a photon of light for each molecule of ATP.

When an aliquot of luciferase–luciferin is added to ATP extracted from a sample of cell suspension, the light generated can be detected in a bioluminometer.

• The resulting signal is amplified and then expressed as a digital or analogue data output

ATP bioluminometry is most suitable for direct measurement of samples that are not coloured, as quenching of light may be a problem.

• However, as the method is very sensitive, samples may require considerable dilution, which often overcomes any colour quenching problem

28. Write a note in ascomycetes..5

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Asexual reproduction:
• In asexual reproduction, the tips of specialized hyphae (conidiophores) form chains of haploid conidiospores that are usually wind dispersed.
• **Sexual reproduction:**
• Sexual spores or ascospores are contained within a sac-like ascus, which may be enclosed within a fruiting body or ascocarp.

The yeasts, such as saccharomyces cerevisiae, produce the equivalent of an ascus during sexual reproduction and most bud during asexual reproduction in a manner similar to the formation of conidiospores

29. What are Antibiotics and their effect on microbes..5

Antibiotics may have cidal or static effects on a range of microorganisms.
• They are classified according to their range of effectiveness and the general microbial group they act against, i.E. Antibacterial, antifungal or antiprotozoal.
• The range of bacteria or other microorganisms affected by a specific antibiotic is expressed as its spectrum of action.

Antibiotics effective against a wide range of gram-positive and gramnegative bacteria are said to be broad spectrum.
• If they are predominantly effective against gram-positive or gram-negative bacteria, they are narrow spectrum
• Limited spectrum antibiotics are effective against only a single organism or disease.

Antibiotics are low molecular weight compounds that are produced as secondary metabolites by mainly soil microorganisms.
• They are non-protein molecules, although there are some peptide antibiotics.
• Most producer microorganisms form a spore or other dormant cell.

Among the filamentous fungi, the notable antibiotic producers are penicillium and cephalosporium (acremonium), which are the main source of b-lactam antibiotics, e.G. Penicillin and related compounds.

• Within the bacteria, the filamentous actinomycetes, especially many streptomyces species, produce a variety of antibiotics, including aminoglycosides, macrolides and tetracyclines.